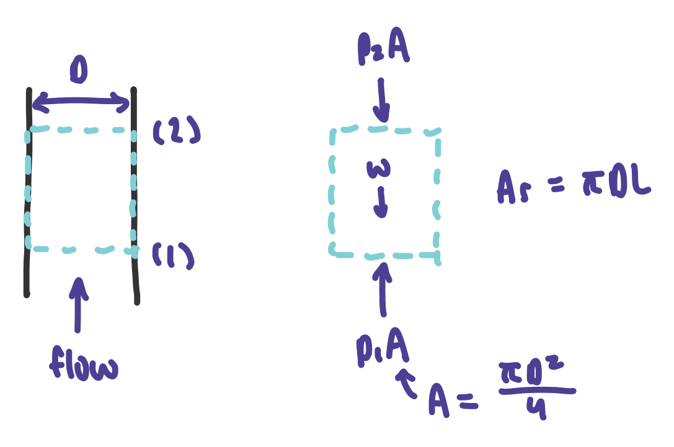
Control volume

Sunday, December 14, 2025 2:19 PM

Control volume

Ex. flow in vertical pipe

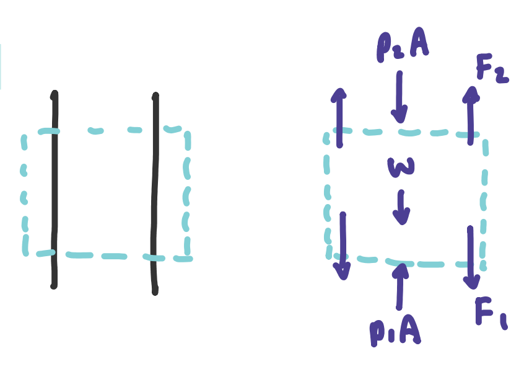
method 1:


D
(2)
(1)
flow
p_2A
w
p_1A
$A = \frac{\pi D^2}{4}$

$A_s = \pi DL$

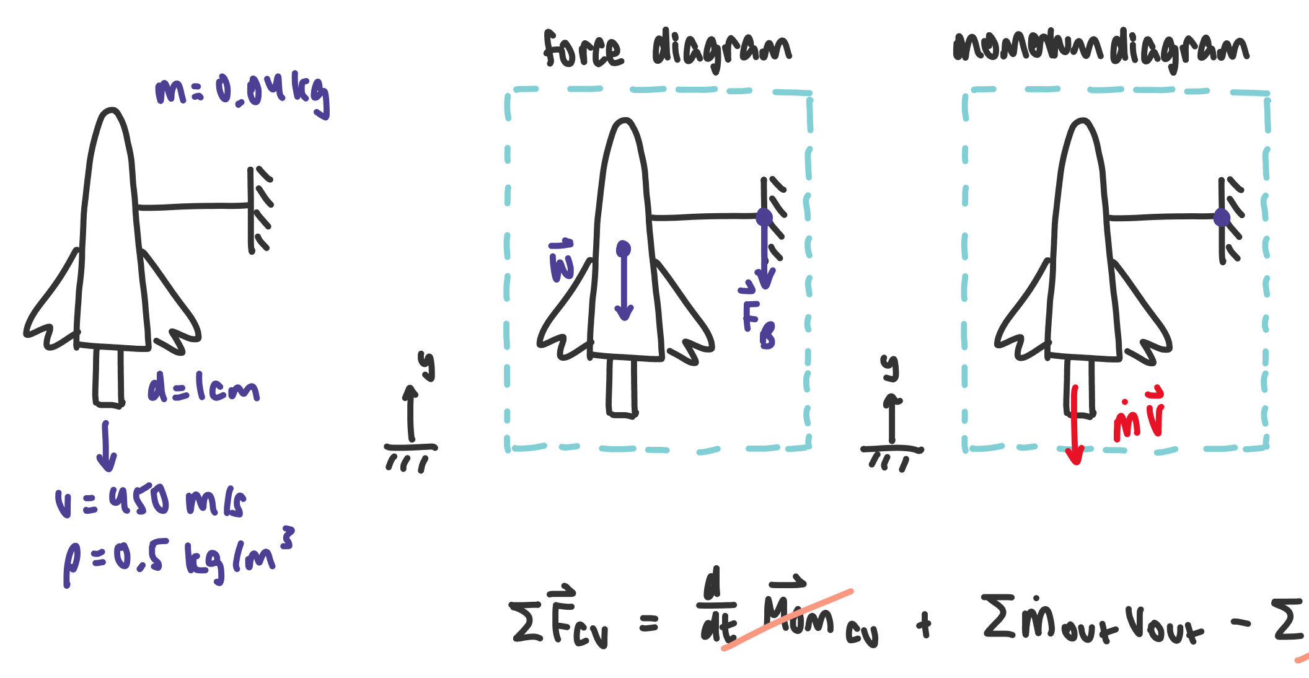
$$\Sigma F_{cv} = [\underbrace{(p_1-p_2)\frac{\pi D^2}{4}}_{\text{pressure}} - \underbrace{\tau\pi DL}_{\text{shear}} - \underbrace{\rho g\frac{\pi D^2}{4}L}_{\text{weight}}]\vec{k}$$

method 2:


p_2A
F_2
w
p_1A
F_1

$$\Sigma F_{cv} = [\underbrace{(p_1-p_2)\frac{\pi D^2}{4}}_{\text{pressure}} + \underbrace{F_2-F_1}_{\text{pipe wall}} - (\underbrace{\rho g\frac{\pi D^2}{4}L}_{\text{weight of fluid}} + \underbrace{W_p}_{\text{weight of pipe in CV}})]\vec{k}$$

Ex.


force diagram
momentum diagram
m = 0.04 kg
d = 1cm
v = 450 m/s
ρ = 0.5 kg/m³
y
$\vec{W}$
$\vec{F}_B$
y
$\dot{m}\vec{V}$

① stationary CV

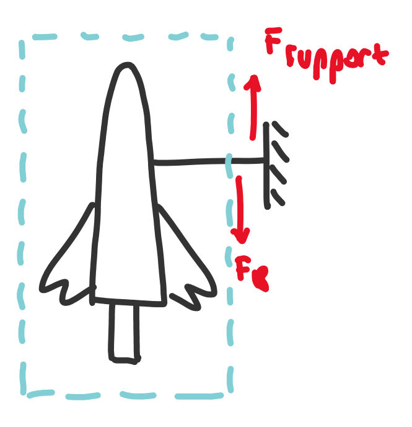
② stationary reference frame

$$\Sigma\vec{F}_{cv} = \cancel{\frac{d}{dt}\vec{M}om_{cv}} + \Sigma\dot{m}_{out}V_{out} - \cancel{\Sigma\dot{m}_{in}V_{in}}$$

$$(-W-F_B)\vec{j} = (\rho VA)(-V\vec{j})$$

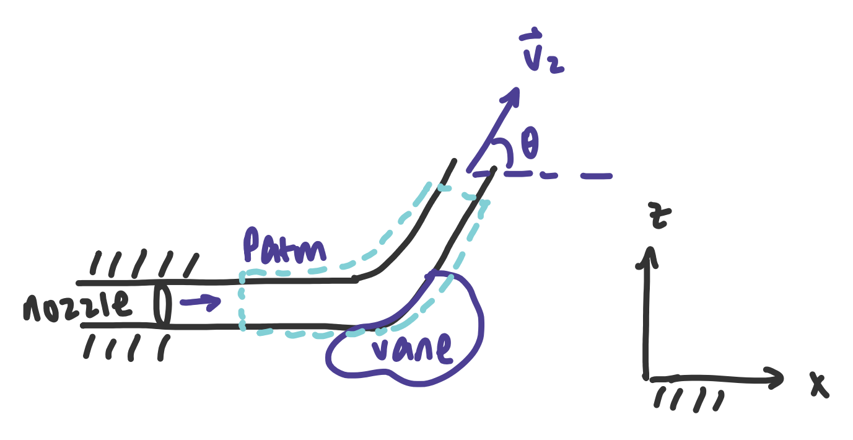
$$F_B = (0.05)(450)^2(\tfrac{1}{4}\pi)(0.01)^2 - (0.04)(9.81)$$

$$= 7.56\text{ N} > 0 \quad \Rightarrow \quad \vec{F}_B = -7.56\vec{j}\text{ N}$$


$F_{support}$
F_B

$\vec{F}_{support} = -\vec{F}_b = 7.56\vec{j}$ N (upward)

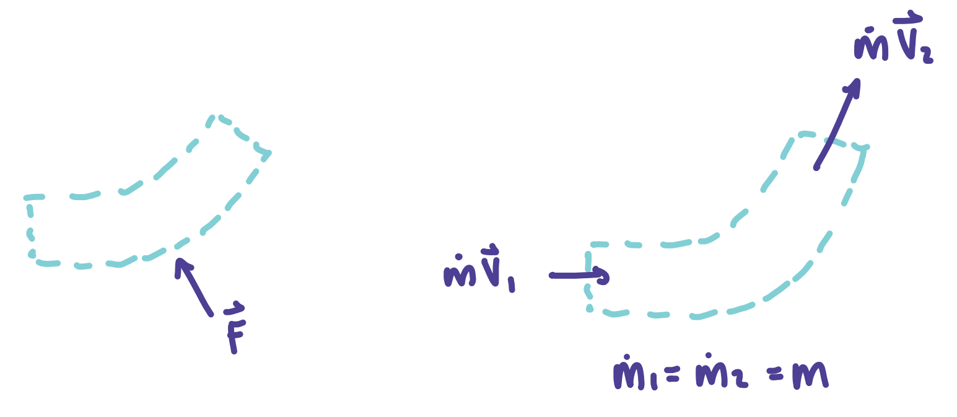
Ex. water jet striking vane


$\vec{V}_2$
θ
P_{atm}
nozzle
vane
z
x

① stationary CV

② stationary ref. frame

force diagram: momentum diagram:


$\vec{F}$
$\dot{m}\vec{V}_2$
$\dot{m}\vec{V}_1$
$\dot{m}_1 = \dot{m}_2 = m$

$$\Sigma\vec{F}_{cv} = \dot{m}(\vec{V}_2 - \vec{V}_1)$$

$$= \dot{m}(V_2\cos\theta\vec{i} + V_2\sin\theta\vec{k} - V_1\vec{i})$$

Using Bernoulli,

$$\frac{p_1}{\rho} + \frac{V_1^2}{2} + gz_1 = \frac{p_2}{\rho} + \frac{V_2^2}{2} + gz_2$$

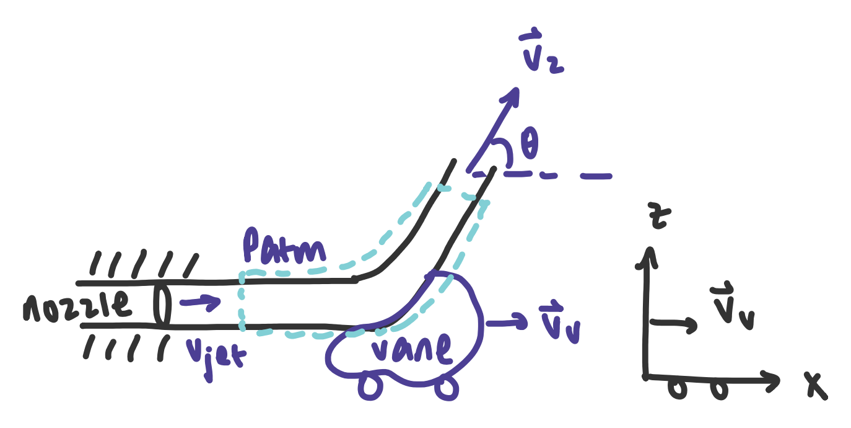
$$p_1 = p_2 = P_{atm},\quad z_1 \approx z_2 \quad\Rightarrow\quad V_1 = V_2 = V$$

$$\vec{F}_{cv} = (\rho VA)(V\cos\theta\vec{i} + V\sin\theta\vec{k} - V\vec{i})$$

$$= \rho V^2A[(\cos\theta - 1)\vec{i} + \sin\theta\vec{k}]$$

$$\vec{F}_{cv} = -\vec{F}_{vane} = \vec{F}_{\text{required to keep vane stationary}}$$

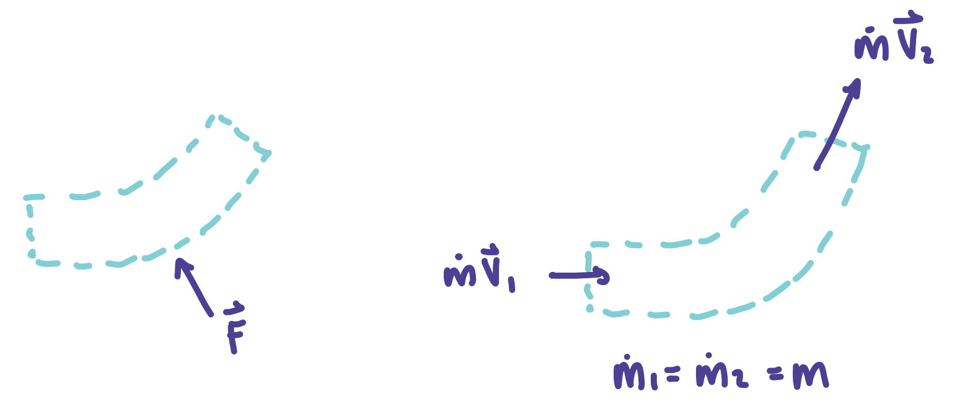
Ex. suppose vane above is moving


$\vec{V}_2$
θ
P_{atm}
nozzle
V_{jet}
vane
$\vec{V}_v$
z
$\vec{V}_v$
x

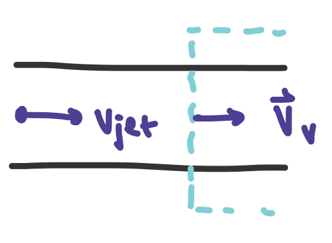
① moving CV

② moving RF with CV

force diagram: momentum diagram:


$\vec{F}$
$\dot{m}\vec{V}_2$
$\dot{m}\vec{V}_1$
$\dot{m}_1 = \dot{m}_2 = m$

$$\vec{F}_{cv} = \underset{\text{wrt CS}}{\dot{m}}(\vec{V}_2 - \underset{\text{wrt RF}}{\vec{V}_1})$$



$$\dot{m} = \rho(V_{jet} - V_v)A_1$$

(as $\vec{V}_{jet} \to \vec{V}_v$, $\dot{m} \to 0$)

$\vec{V}_1 = (\vec{V}_{jet} - \vec{V}_v)\vec{i}$ $\quad\vec{V}_2 = ?$

Bernoulli:

$$\cancel{\frac{p_1}{\rho}} + \frac{V_1^2}{2} + \cancel{gz_1} = \cancel{\frac{p_2}{\rho}} + \frac{V_2^2}{2} + \cancel{gz_2}$$

$$\|V_1\| = \|V_2\|$$

$$\vec{V}_2 = V\cos\theta\vec{i} + V\sin\theta\vec{k}$$

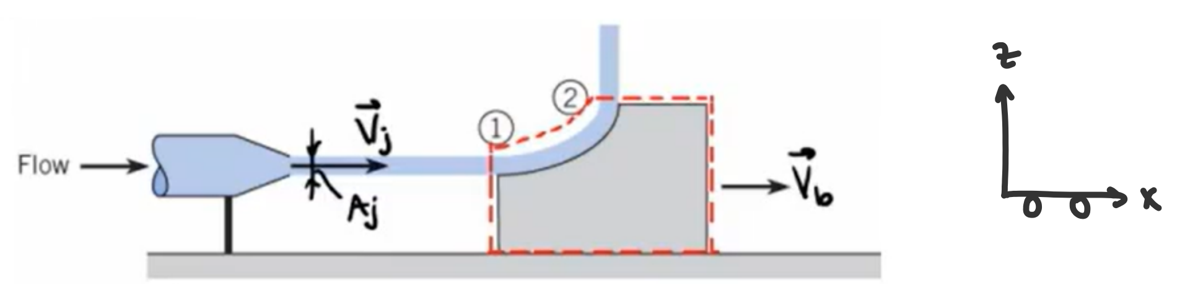
$$= (V_{jet}\ V_v)\cos\theta\vec{i} + (V_{jet} - V_v)\sin\theta\vec{k}$$

$$\vec{F}_{cv} = \underbrace{\rho(V_{jet} - V_v)A_1}_{\dot{m}}[\underbrace{(V_{jet} - V_v)\cos\theta\vec{i} + (V_{jet} - V_v)\sin\theta\vec{k}}_{\vec{V}_2} - \underbrace{(V_{jet} - V_v)\vec{i}}_{\vec{V}_1}]$$

$$= \rho(V_{jet} - V_v)^2A_1[(\cos\theta - 1)\vec{i} + \sin\theta\vec{k}]$$

$$\vec{F}_{vane} = -\vec{F}_{cv}$$

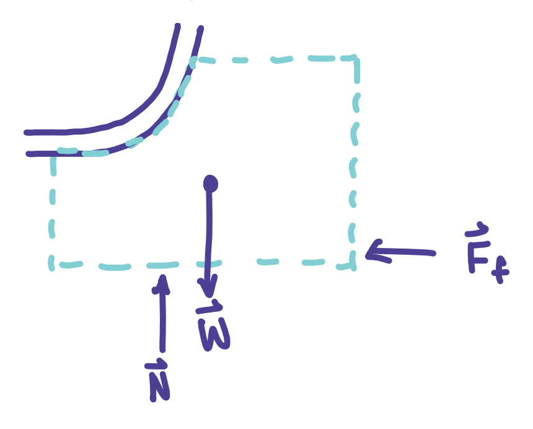
Ex. jet with moving block deflected


Flow
$\vec{V}_j$
A_j
1
2
$\vec{V}_b$
z
x

① moving CV with $\vec{V}_b$

② moving RF with CV

force diagram


$\vec{F}_f$
$\vec{W}$
$\vec{N}$

$$\Sigma\vec{F}_{cv} = \dot{m}(\vec{V}_2 - \vec{V}_1)$$

$$= \rho A_j(V_j - V_b)$$

(if $V_b \to V_j$ then $\dot{m} \to 0$)

$$\vec{V}_1 = (V_j - V_b)\vec{i}$$

$$\vec{V}_2 = (V_j - V_b)\vec{k}$$

$\|\vec{V}_1\| = \|\vec{V}_2\|$ from Bernoulli

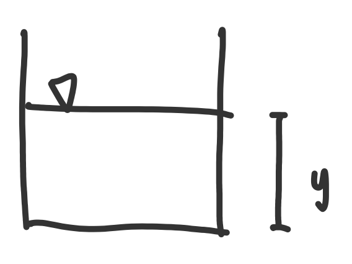
$$\Sigma\vec{F}_{cv} = -F_f\vec{i} + (N - W)\vec{k}$$

$$\underbrace{-F_f\vec{i} + (N-W)\vec{k}}_{\Sigma\vec{F}_{cv}} = \underbrace{\rho A_j(V_j - V_b)}_{\dot{m}}[\underbrace{(V_j - V_b)\vec{k} + V_b\vec{i}}_{\vec{V}_2} - \underbrace{V_j\vec{i}}_{\vec{V}_1}]$$

Open-channel flow

Froude number

$$Fr = \frac{\overset{\leftarrow \text{fluid}}{V}}{\underset{\uparrow\text{wave}}{\sqrt{gy}}}$$



$Fr < 1$: subcritical

$Fr = 1$: critical

$Fr > 1$: supercritical

Compressible flow

If density varies, Bernoulli cannot be used

Mach number

$$M = \frac{V}{c}$$

← flow velocity

← speed of sound in fluid

$M_\infty \leq 0.3$: incompressible

$M_\infty > 0.3$: compressible

$M_\infty < 1$: subsonic

$M_\infty = 1$: sonic

$M_\infty > 1$: supersonic

subsonic : A↑ V↓

A↓ V↑

supersonic : A↑ V↑

A↓ V↓